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Exp-1

STUDY OF VARIOUS ~~LINUX~~ NETWORK COMMANDS USED IN LINUX AND WINDOWS

1) arp -a : Shows IP address of your computer along with IP address & MAC address of your router

O/P:

Interface: 192.168.156.1 --- 0x7

Internet address	Physical address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-92	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

2) hostname:- Displays name of your computer.

O/P:

DESKTOP-MFFKGEN

3) ipconfig/all:- Displays detailed configuration information about TCP/IP connection and type of ethernet adapter in the system.

O/P:-

Windows IP Configuration

Host Name . . . . . : DESKTOP-MFFKGEN
Primary DNS suffix . . . :
Node type . . . . . : Mixed
IP Routing enabled . . . . . : No
WINS proxy enabled . . . . : No

~~Ethernet adapter Ethernet~~

Ethernet adapter ethernet:

Media state ..... : Media disconnected

Connection-specific DNS Suffix ... :

Description ..... : Realtek PCIe GbE Family Controller

Physical Address..... : C4-D6-D3-A2-72-D3

DHCP enabled..... : Yes

Autoconfiguration Enabled.... : Yes.

4) nbtstat -a:- Displays protocol statistics and current TCP/IP connections using NBT.

O/P:-

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

-a (adapter status) -r (resolved) -RR (ReleaseRefresh)

-A (Adapter status) -R (reload)

-c (cache) -S(Sessions)

-n (names) -s (sessions)

5) netstat:- Displays a variety of statistics about a computer active TCP/IP connections

O/P:

Active Connections

Proto	Local address	Foreign address	State
TCP	10.76.96.60:49175	10.76.96.163:domain	TIME_WAIT
TCP	10.76.96.60:52239	142.250.182.3:http	ESTABLISHED
TCP	10.76.96.60.56954	10.76.96163:domain	TIME.WAIT

6) nslookup:- Display DNS details such as IP Address of the particular computer

O/P:-

Server: Unknown
Address: 10.76.96.163

7) Pathping:- Traces the root to destination address then launches a 25 second test of each router along the way.

O/P:-

pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
[-P period] [-q num_queries] [-w timeout]
[-4] [-6] target_name

LINUX NETWORKING COMMANDS

1) ip:- Helps from setting up new system and assigning IPs to troubleshooting existing systems.

2) ifconfig:- The ifconfig command is a staple in many sysadmin's tool belt for configuring and troubleshooting networks.

3) mtr:- MTR is a program with CLI that serves as a network diagnostic and troubleshooting tool.

4) tcpdump:- The tcpdump command is designed for capturing and displaying packets

5) Ping:- Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request Message.

~~ip~~ o/p:-

Usage:

1) ip → o/p:-

Usage: ip [OPTIONS] OBJECT {COMMAND | help}

ip [-force] -batch filename

where OBJECT := {link | address | addrlabel | route | rule | neigh | ntable | tunnel | tuntap | maddress | mroute | mrule}

OPTIONS := {-V[ersion] | -s[tatistics] | -d[etails] | -r[esolve | -h[uman-readable] | -iec}

2) ifconfig → o/p:-

lo: flags=73<UP, LOOKBACK, RUNNING> mtu 65536

inet 127.0.0.1 netmask 255.0.0.0

inet6 ::1 prefixlen 128 scopeid 0x10<host>

loop txqueuelen 1000 (Local Loopback)

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 12329 bytes 811650 (792.16 KiB)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

3) mtr → o/p:-

My traceover [v0.87]

localhost.localdomain (::)

Keys: Help Display mode Restart statistics Order of fields quit

Host	Packets Loss%	Snt	Pings Last	Avg	Best	Wrst	StDev
1. ::1	0.0%	185	0.2	0.2	0.1	0.5	0.0

4) tcpdump - O/P:-

tcpdump: enp350 You dont have permission to capture on that device

5) ping - O/P:-

Usage ping: [-aAbBdDfhLnOqrRUvV] [-c count] [-i interval]
[-I interface] [-m mark] [-M pmtudisc] [-l preload] [-p pattern]
[-Q tos] [-S sndbuf] [-t ttl]
[-T timestamp-option]

Ping google.com

O/P

PING google.com (142.250.190.46) 56(84) bytes of data
64 bytes from lga25s71-in-f14.1e100.net (142.250.190.46): icmp_seq=1 ttl=116
64 bytes from lga25s71-in-f14.1e100.net (142.250.190.46): icmp_seq=2 ttl=116
^C
--- google.com ping statistics ---
2 pkt transmitted, 2 received, 0% pkt loss, time 3004ms

RESULT:-

Thus the various command network used in linux and windows are executed successfully.